

SynthProof — safe "stand-in" data you can actually trust

Hospitals and banks hold sensitive records. They want to share data with researchers **without exposing real people**. SynthProof helps them share safe fake data — and, for the first time, lets anyone **check that it's really safe** instead of just taking their word for it.

The story, in plain English



1. The need.

A hospital wants researchers to study its patients — but it can't hand over real medical records. That would expose real people.



2. The usual fix: "fake" data.

So they create **stand-in data** — a realistic look-alike that behaves like the real records but isn't any actual person. Like a crash-test dummy standing in for a real human.



3. The hidden problem.

While making that fake data, they often **leave clues behind by accident** — a bit like handing out a photocopy but forgetting your fingerprint is on it. A clever snoop could use those clues to figure out real people. And today, **nobody checks** for these leftover clues, and there's **no way to prove** the fake data is safe — you just have to trust whoever made it.

What we built



A safety inspector

Software that reads the released data package and

flags every leftover clue

— like an airport scanner for fake data.



A tamper-proof label

A "nutrition label" for the data with a

digital wax seal

. Anyone can check the seal — so you don't have to just trust the maker.



A one-click check

Anyone who receives the data can

press one button

and see, from the package alone, whether it's safe.

Does it actually work? Yes.

The fake data is about 9 out of 10 as useful as the real data — with no way to tell who was really in it.

We tested on four real datasets (hospital patients, bank customers, and two census sets — **every single row**). A computer that learns from our safe fake data is nearly as smart as one that learned from the real, private data — **but a snoop trying to tell if you were in it does no better than flipping a coin**. So it's useful **and** safe at the same time.

Why this is better than what others did

- ✓ Some researchers **proved the leaks exist** — but they only pointed at the problem. **We built the tool that finds and blocks them.**
- ✓ One team **drew up a plan** for a safety label — but never actually built it or made it tamper-proof. **We built it, sealed it, and made it checkable.**
- ✓ We checked **290 fake datasets shared online** and **12 releases from big names** (Apple, Google, the US Census...). **Not one** comes with a way to prove it's safe. **Ours does.**

In one sentence: everyone else either makes fake data *or* points out its hidden leaks — **SynthProof is the first to hand you fake data together with a sealed, checkable proof that it's safe**, so you can verify it yourself instead of just hoping.